

BBC News Articles

Topic Modeling and Text Clustering Analysis

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Abstract

This project explores the discovery of latent thematic structures within a large-scale corpus of 35,860 BBC News articles spanning the 2013-2024 period. Given the constraint of working exclusively with short-form text (abstracts averaging ≈ 17 words), we implemented a robust text mining pipeline to overcome high sparsity and lexical overlap. Our methodology integrates unsupervised Text Clustering (K-Means) with probabilistic and algebraic Topic Modeling (LDA and NMF). Through a comparative evaluation using Silhouette, Davies-Bouldin, and Calinski-Harabasz metrics, we demonstrate that K-Means effectively partitions the corpus into distinct events, while NMF provides the highest lexical purity for thematic labeling. The results successfully isolate high-intensity news niches—such as the Ukraine-Russia and Israel-Gaza conflicts—from general journalistic prose.

Keywords: Text Mining, BBC News, K-Means, LDA, NMF, Short-Text Analysis.

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1 Introduction

In the era of information overload, the ability to automatically categorize and extract meaning from vast streams of journalistic content is essential for news organizations and media researchers alike. This project analyzes a substantial dataset provided by BBC News, consisting of 35,860 articles published between 2013 and 2024, with a predominant focus on recent coverage from 2022 to 2024.

A significant challenge inherent in this study is the nature of the data: our input consists solely of short text descriptions or abstracts, typically averaging 17 words per entry, rather than the full bodies of the articles. This constraint introduces high levels of data sparsity, making traditional frequency-based analysis difficult. To address this, we developed a comprehensive Text Mining pipeline designed to transform these brief representations into a high-dimensional vector space using TF-IDF and Count Vectorization.

The core of our research focuses on an integrated approach to unsupervised content understanding through two primary methodologies:

1. **Text Clustering:** Utilizing the K-Means algorithm to identify hard partitions and natural groupings within the data, validated against alternative models like Hierarchical Clustering and Gaussian Mixture Models (GMM).
2. **Topic Modeling:** Employing Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF) to discover the "soft" thematic mixtures that define the BBC's editorial landscape.

By benchmarking these algorithms, we aim to justify the most effective model for short-text analysis and provide actionable insights into news categorization, trend detection, and archive organization. The following sections detail our preprocessing strategy, the optimization of our models, and a comparative evaluation of the identified thematic structures.

2 Environment Setup

The computational environment for this project was designed to handle the complexities of Natural Language Processing (NLP) and unsupervised machine learning. The analysis was conducted within a cloud-based Python environment (Google Colab), ensuring reproducibility and access to high-performance computing resources. The setup is divided into four main functional blocks:

2.1 Data Management and Linear Algebra

For the structural handling of the BBC News corpus, **Pandas** was utilized for its robust DataFrame structures,

while **NumPy** provided the necessary support for high-performance multi-dimensional array operations. These libraries are fundamental for managing the metadata associated with the short text descriptions.

2.2 Natural Language Processing (NLP)

The linguistic pipeline was primarily supported by the **Natural Language Toolkit (NLTK)**. This included:

- **Tokenization:** The fundamental process of breaking down the raw text strings into discrete units (tokens). We utilized the *Punkt* tokenizer to accurately segment the article descriptions into individual words and punctuation marks, creating the initial structure for analysis.
- **Lexical Normalization:** A vocabulary consolidation technique used to reduce inflectional forms to a common base. We employed a dual approach: *WordNetLemmatizer* to resolve words to their dictionary root, lemma, based on context, and *PorterStemmer* to algorithmically strip suffixes, ensuring that variations like "politics" and "political" are analyzed as a single semantic entity.
- **Stop-words Filtering:** Removal of high-frequency words that lack significant semantic value in a news context. This step is crucial for short-text analysis, as it reduces noise and forces the model to focus on the content-bearing keywords unique to each news topic.
- **POS Tagging:** The algorithmic assignment of grammatical categories to each token using the *averaged_perceptron_tagger*. This syntactic labeling aids in disambiguation and helps refine the lemmatization process by providing necessary context.

2.3 Machine Learning and Dimensionality Reduction

The core of the analysis relies on **Scikit-Learn**.

- **Vectorization:** the process of converting *unstructured* text data into a *structured* numerical format (vectors) that machine learning algorithms can process. **TfidfVectorizer** and **CountVectorizer** were used to transform text into a numerical format suitable for modeling.
- **Unsupervised Models:** Implementation of **Non-Negative Matrix Factorization (NMF)** for topic extraction and **K-Means** for document clustering.
- **Validation Metrics:** Tools such as the *Silhouette Score* and *Davies-Bouldin Index* were integrated to evaluate the cohesion and separation of the clusters.

2.4 Visualization and Reporting

To interpret the high-dimensional data, **Seaborn** and **Matplotlib** were configured with standardized aesthetic settings (whitegrid style). Furthermore, the **WordCloud** library was utilized to provide an intuitive visual representation of the most prominent terms within each identified topic.

3 Dataset Overview

The dataset serves as the foundational corpus for this study, consisting of articles retrieved from the **BBC News platform**. This section details the structural composition, the intrinsic quality of the data, and the specific characteristics that define the scope of the Text Mining task.

3.1 Data Source and Temporal Scope

The corpus was programmatically acquired via the **kagglehub** API, targeting the *BBC News Articles* dataset. It encompasses a continuous archive of news stories published between **August 2013 and July 2024**. This time frame is particularly significant as it covers major global events, ranging from geopolitical conflicts to economic shifts, ensuring a high degree of thematic diversity (high entropy) within the text.

3.2 Feature Engineering and Variable Analysis

The raw dataset is structured as a tabular collection of 35,860 entries and 5 primary features. While the metadata provides context, the focus of the unsupervised pipeline is the **description** field.

Feature	Analytical Utility
title	Provides concise semantic anchors for identifying the primary subject of the article.
pubDate	Enables temporal segmentation and trend analysis over the 2013-2024 period.
guid	Serves as a unique identifier for data integrity and duplicate tracking.
link	Reference URL for the full-text source (not utilized in this vectorization).
description	The study's primary target; contains a short-text summary of the event.

Table 1: Detailed Variable Description and Utility

3.3 Data Integrity and Quality Assessment

A rigorous inspection was conducted to evaluate the dataset's readiness for the NMF and K-Means algorithms. The findings indicate an exceptionally high level of data quality:

- **Completeness:** The missing value analysis yielded a **0% null-rate** across all features. In Text Mining, this is critical as it prevents bias or the need for imputation strategies that could distort the document-term matrix.
- **Redundancy Control:** Despite the high volume of articles, the dataset contains **zero exact duplicate rows**. This ensures that the clustering algorithm is not biased toward specific repeating entries, allowing for a more balanced distribution of centroids in the K-Means process.
- **Computational Efficiency:** With a memory footprint of **19.36 MB**, the dataset allows for rapid iteration and testing of hyperparameters without the need for distributed computing frameworks, facilitating a more granular optimization of the NMF decomposition.

3.4 The Short-Text Constraint

A key characteristic of this dataset is the brevity of the **description** field. Initial inspection shows that these summaries are highly distilled (approx. 17–20 words per entry). Unlike full-text corpora, this "short-text" nature presents a challenge for traditional clustering due to the **sparsity of the vector space**.

This observation justifies our methodological choice: using **NMF** as a dimensionality reduction step before clustering. By mapping these sparse descriptions into a dense "topic space," we can capture semantic similarities that would be missed by a simple word-count comparison. The statistical uniqueness of the descriptions (33,124 unique summaries) further confirms that the corpus provides a sufficiently wide vocabulary for meaningful topic extraction.

4 Explorative Data Analysis

This phase focuses on understanding the temporal dynamics and structural complexity of the BBC News corpus. Identifying these patterns is essential for calibrating the subsequent vectorization and unsupervised modeling steps.

4.1 Temporal Patterns and Publishing Trends

To evaluate the longitudinal composition of the dataset, the **pubDate** field was converted into a structured datetime

format. The corpus spans a total of **3690 days**, ranging from **August 30, 2013**, to **July 3, 2024**.

Findings and Interpretation

The temporal analysis, visualized through distribution plots, reveals several insights into the editorial cycle and dataset coverage:

- **Yearly Volume:** The publication frequency exhibited a clear peak in **2023**, with the dataset containing over 14,000 entries for that year. In comparison, 2022 maintained a high volume of approximately 12,000 articles, establishing a robust baseline for the analysis.
- **The Weekly Cycle:** A distinct "weekend drop" is observable in the editorial patterns. While publishing activity remains high and consistent from Monday to Friday (averaging approx. 5,400 articles per day), there is a reduction of roughly **20% on Saturdays and Sundays**. This cyclic fluctuation confirms the structured editorial schedule typical of a major news organization.

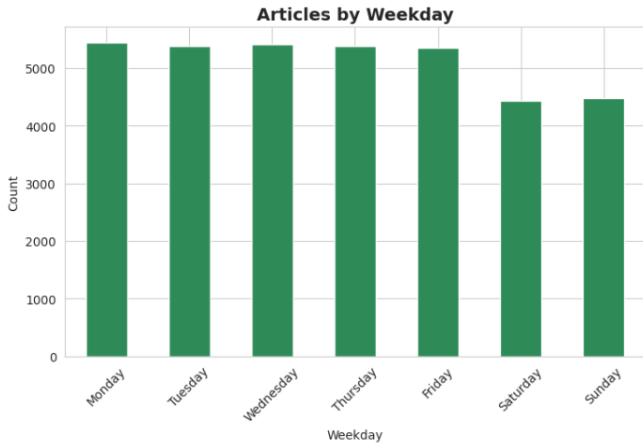


Figure 1: **Temporal distribution of the BBC News corpus.** The charts illustrate the publishing volume across years, months, and weekdays. Notably, the *Articles by Weekday* plot reveals a consistent editorial output from Monday to Friday, followed by a discernible reduction during the weekend, reflecting standard newsroom cycles. The monthly and yearly trends highlight the longitudinal consistency of the dataset.

- **Timeline Volatility:** The monthly timeline illustrates a significant surge in article density in early 2022. This period coincides with the **Russian invasion of Ukraine**, a major global event that dominated international news cycles and is expected to emerge as a primary latent topic in the NMF model.

4.2 Text Complexity and Length Analysis

For this study, a "**Semantic Unit**" was defined by concatenating the **title** and the **description** fields. This approach ensures that the context provided by the headline is preserved alongside the summary, creating a robust input for the vectorization process.

Statistical Profile

The combined text was analyzed to evaluate information density and token distribution. The results are summarized in the table below:

Metric	Character Length	Word Count
Average	165.16	27.27
Median	159.00	26.00
Maximum	332	58
Minimum	36	5

Table 2: Descriptive Statistics for Combined Text Length

Implications for Text Mining

The distribution analysis (visualized in the histograms) shows a near-normal distribution with a slight right skew, leading to the following strategic conclusions for the pipeline:

1. **Information Sparsity:** With a median of **26 words**, these documents are significantly shorter than full-length news articles. This brevity increases the risk of **feature sparsity** in the TF-IDF matrix. Consequently, our preprocessing must prioritize lemmatization over aggressive stemming to preserve semantic richness.
2. **Outlier Management:**
 - **Short Articles (< 10 words):** Entries with as few as 5 words likely represent breaking news alerts or "live" updates. These may lack sufficient "signal" for the clustering algorithm and are candidates for removal if they create noisy clusters.
 - **Long Articles (> 45 words):** The maximum length is 58 words, which is still relatively concise. Unlike full-text datasets where long documents can dominate TF-IDF weights, this corpus is naturally balanced, favoring more stable K-Means centroids.
3. **TF-IDF Sensitivity:** Because lengths are fairly uniform, TF-IDF normalization will not need to aggressively penalize document length, allowing the model to focus strictly on the discriminative power of specific news-related keywords.

The synergy between temporal spikes (e.g., the 2022 surge) and consistent text length suggests that while the **topics** vary greatly over time, the **format** of the reporting remains stable, providing an ideal environment for unsupervised semantic discovery.

5 Text Pre-processing Pipeline

The transition from raw natural language to a structured computational format is the most critical phase in the analytical pipeline. For the BBC News corpus, preprocessing is not merely a cleaning task but a strategic process of **semantic dimensionality reduction**. Given that the news descriptions are inherently brief, with an initial average of ≈ 27 words, every token removed or normalized profoundly impacts the density and quality of the subsequent vector space.

5.1 Methodological Rationale and Pipeline Design

Our preprocessing architecture was engineered to transform noisy, inconsistent raw text into a normalized, content-bearing format. The pipeline is designed as a modular sequence of four specific functions to maximize semantic intersection across the corpus.

- **Case-Insensitive Normalization and Regex Cleaning:** The function `clean_text()` handles the removal of non-semantic artifacts. All text is converted to lowercase to ensure that "Ukraine", "ukraine", and "UKRAINE" are mapped to a single token, drastically reducing vocabulary size.

- *URL, Email, and HTML Removal:* Using Regular Expressions (Regex), we eliminate web-scraped residuals (e.g., `http\S+, <.*?>`). These elements act as statistical outliers and carry no thematic value.
- *Special Characters and Digits:* We strip punctuation and numbers to focus exclusively on alphabetic descriptors. While dates carry temporal context, in a bag-of-words approach, they fragment the feature space.

- **Stop-word Filtering and Length Constraints:** The `remove_stopwords()` function utilizes the *NLTK* English corpus to filter high-frequency functional words (e.g., "the", "and", "is"). Furthermore, we implemented a **minimum character length filter** (> 2). This ensures that short noise particles are removed while preserving meaningful terms like "war" or "win", concentrating the TF-IDF weight on discriminative nouns and verbs.

- **Lemmatization: The Choice of Precision:** It can be defined as the linguistic process of reducing words

to their base or dictionary form (lemma) by considering their context and grammatical role. A core design decision was the implementation of **WordNet Lemmatization** instead of heuristic **Stemming**.

- *Stemming:* Uses crude heuristic processes to chop off the ends of words (e.g., "studies" $\rightarrow$ "studi"). This often results in non-dictionary roots, which is detrimental to the readability of cluster labels.
- *Lemmatization:* Uses a lexical database to map words to their actual dictionary root (lemma) based on morphological analysis (e.g., "running" $\rightarrow$ "run", "was" $\rightarrow$ "be").

Given the brevity of BBC News snippets, we prioritized accuracy over computational speed. Lemmatization ensures that "vowed", "vows", and "vowing" are consolidated into the correct base form ("vow"), preventing vocabulary inflation and creating denser feature vectors.

5.2 Impact Analysis and Statistical Distillation

The application of this pipeline to the 35,860 articles resulted in a significant refinement of the corpus. The process flow involved the removal of empty or excessively short texts (< 10 characters post-cleaning) to maintain high data integrity.

Applying the modular pipeline to the 35,860 articles resulted in a structural transformation of the dataset. The process flow involved a rigorous validation step where any text reduced to fewer than 10 characters post-cleaning was discarded to maintain high data integrity. Interestingly, the final dataset retained all 35,860 articles, demonstrating that the editorial quality of the BBC provides a resilient "semantic skeleton" that remains consistently informative even after stripping noise.

The quantitative impact of this distillation is significant: the average word count per article was reduced from **27.27 to 17.53**, with the median word count shifting from **26.00 to 17.00**. This reduction is not a loss of information but rather a **concentration of meaning**. By removing approximately 35% of the total text—primarily grammatical "glue" and redundant inflections—the corpus was transformed from a noisy collection of sentences into a high-density vocabulary of approximately 8,000 to 12,000 unique content-bearing tokens.

5.3 Preprocessing Results and Semantic Alignment

The qualitative success of the pipeline is best observed through direct comparison. For example, a raw headline such as "Ukraine: Angry Zelensky vows to punish Russian atrocities..." was successfully distilled into "ukraine angry zelensky vow punish russian atrocity...".

By consolidating various morphological forms (e.g., "vows" and "vowing" to "vow") and removing non-discriminative terms, we have created a focused feature space. This ensures that the subsequent TF-IDF vectorization phase will operate on a manageable and highly relevant vocabulary, leading to more distinct document representations, sharper cluster separation, and more interpretable topics in the LDA and NMF models.

Findings and Observations:

- **Data Integrity:** Zero articles were removed after cleaning, indicating that even the shortest BBC descriptions contain a resilient "semantic skeleton" that remains informative after stripping noise.
- **Lexical Distillation:** The reduction from 27.27 to 17.53 average words represents a **35% distillation of meaning**. By removing "grammatical glue," we have ensured that the TF-IDF vectorization (the next stage) will be based purely on terms with high discriminative power.
- **Sample Transformation:** As seen in the transition from *"Ukraine: Angry Zelensky vows..."* to *"ukraine angry zelensky vow..."*, the text has been stripped to its essential thematic components, facilitating more coherent cluster formation and clearer topic modeling.

6 Text Representation

The objective of text representation is to project cleaned linguistic tokens into a high-dimensional numerical vector space. This transformation allows machine learning algorithms to perform mathematical operations on text, where the geometric distance between vectors represents semantic dissimilarity.

6.1 TF-IDF Vectorization Strategy

We employed the **Term Frequency-Inverse Document Frequency (TF-IDF)** weighting scheme. Unlike simple frequency-based methods such as Count Vectorization, TF-IDF evaluates the importance of a word by balancing its local frequency within a document (*TF*) against its global scarcity across the entire corpus (*IDF*). This effectively filters out words that are common but lack discriminative power (e.g., "news") while highlighting terms that characterize specific topics (e.g., "inflation" or "zelensky").

The vectorizer was configured with the following strategic hyperparameters to optimize the feature space:

- **Feature Dimensionality ($max_features=5000$):** The vocabulary was restricted to the top 5,000 most significant terms. This constraint mitigates the curse of dimensionality, ensuring computational efficiency while retaining the most informative content-bearing tokens.

- **Frequency Filtering ($min_df=5$, $max_df=0.7$):**

- $min_df=5$: A token must appear in at least five documents to be included. This threshold effectively filters out typographical errors, noise, and rare outliers that lack statistical significance for topic discovery.
- $max_df=0.7$: Terms appearing in more than 70% of the articles are discarded. This acts as a dynamic stop-word filter, removing ubiquitous domain-specific jargon that, due to its high frequency, lacks discriminative power.

- **N-gram Expansion ($ngram_range=(1, 2)$):** We expanded the lexicon to include both unigrams (single words) and **bigrams** (two-word phrases). This is essential for capturing semantic units that lose meaning when split, such as "world cup," "prime minister," or "ukraine war." Our final feature set consists of approximately 70% unigrams and 30% bigrams.

6.2 Feature Space Analysis and Sparsity

The resulting matrix dimensions ($35,860 \times 5,000$) represent a highly sparse high-dimensional space. We recorded a **Sparsity of 99.73%**, meaning that for any given article, only 0.27% of the possible 5,000 features contain a non-zero value.

This sparsity is a positive quality indicator: it confirms that news articles are specialized and focused on specific terminology rather than utilizing the entire vocabulary, which facilitates better cluster separation in the subsequent phases.

6.3 Interpretation of Top-Ranked Features

By calculating the mean TF-IDF scores across the corpus, we identified the 20 most influential terms. These terms serve as a "semantic profile" of the BBC News corpus between 2022 and 2024:

- **Core Journalistic Language:** The term *"say"* (score: 0.0203) dominates the ranking, reflecting the standard journalistic practice of quoting sources and reporting speech.
- **Geopolitical Focal Points:** The presence of *"world"* (0.0118), *"england"* (0.0114), and *"ukraine"* (0.0106) establishes the primary geographic and political axes of the dataset.
- **Thematic Indicators:** The inclusion of *"war"* (0.0079) and *"police"* (0.0074) highlights the prevalence of conflict and domestic security reporting.
- **Bigram Efficacy:** The appearance of *"world cup"* (0.0072) at position 15 validates our n-gram strategy. This phrase carries a specific cultural weight

that would be lost if treated as two separate tokens, providing a clear marker for sports-related clusters.

- **Political Cycle:** The term "*election*" (0.0063) completes the top 20, representing the recurring political narratives present in the corpus.

The reasonable score range (0.006 – 0.020) and the absence of "gibberish" confirm that our preprocessing was successful and that the representation is primed for modeling.

6.4 Strategic Choice: Why TF-IDF for both Clustering and Topic Modeling?

The selection of TF-IDF as our primary representation was a deliberate choice to ensure compatibility across two different analytical paradigms:

- **Synergy with Clustering (Distance-based):** K-Means relies on *Euclidean Distance* to group similar articles. By using TF-IDF, we ensure that documents are not grouped just because they are long, but because they share **distinctive, high-weight terms**. This creates sharper cluster boundaries and more interpretable centroids.
- **Synergy with Topic Modeling (Frequency-based):** For algorithms like NMF and LDA, TF-IDF acts as a built-in "importance filter." By automatically down-weighting ubiquitous words like "bbc" or "said," the model can focus its mathematical energy on discovering the **thematic nuances** (e.g., "cost of living" vs. "interest rates") that define the latent topics.

In summary, TF-IDF provides the necessary balance of statistical precision and noise reduction, serving as a robust foundation for identifying both hard partitions (Clustering) and overlapping thematic mixtures (Topic Modeling).

7 Unsupervised Text Clustering

After representing the corpus in a high-dimensional vector space, we applied the **K-Means** algorithm to identify natural groupings within the BBC News articles. This unsupervised approach allows us to discover latent thematic structures without relying on pre-existing labels, effectively letting the data "speak for itself."

7.1 Optimization: Determining the Ideal Number of Clusters (k)

A critical challenge in K-Means clustering is selecting the optimal value for k . We performed a heuristic search for $k \in [2, 10]$, evaluating the results through multiple statistical lenses to balance granularity and cohesion.

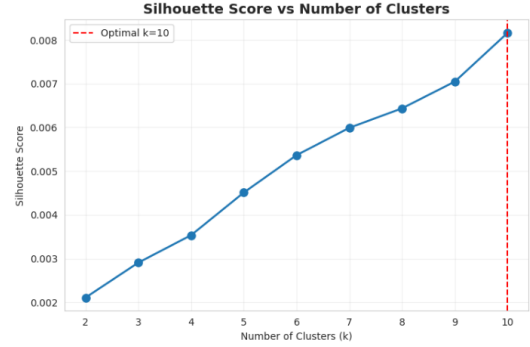


Figure 2: Silhouette Score vs. Number of Clusters (k). The red dashed line indicates the peak at $k = 10$, which represents the highest level of cluster cohesion for this dataset.

As illustrated in the diagnostic plots, we analyzed the following metrics:

- **Silhouette Score:** This was our primary selection criterion. It computes the mean distance between a sample and all other points in the same cluster (cohesion) versus the mean distance to points in the nearest neighboring cluster (separation). Ranging from -1 to 1, a higher value indicates better-defined clusters. The score reached its maximum at $k = 10$ (**0.0080**), justifying the choice of a 10-cluster model.
- **Davies-Bouldin Index:** This metric evaluates the average "similarity" between clusters, where a **lower score** signifies better separation. Our model achieved a value of **8.4238**. While a lower value is theoretically ideal, this result reflects the high-dimensional overlap inherent in semantic data.
- **Calinski-Harabasz Score:** Also known as the Variance Ratio Criterion, this measures the ratio of the sum of between-clusters dispersion and within-cluster dispersion. A **higher score** indicates dense and well-separated clusters. The final score of **96.54** confirms that the model successfully captured structural variance despite the fluid boundaries of news topics.

7.2 K-Means Execution and Population Analysis

The final model was executed using **Lloyd's algorithm** with *k-means++* initialization to ensure stability. Lloyd's algorithm is the standard iterative method used to perform k-means clustering. It partitions a dataset into k distinct groups (clusters) by minimizing the distance between data points and the center of their assigned group. The distribution of the 35,860 articles reveals a highly non-uniform landscape, as shown in the following figure:

The population analysis (Figure 3) provides deep insight into the nature of the corpus, with two clusters playing a pivotal role in the global architecture of the data:

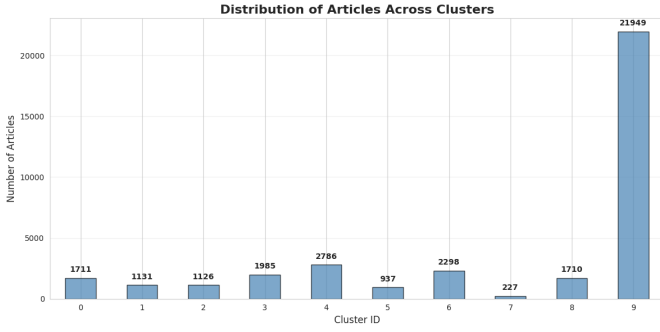


Figure 3: Distribution of the 35,860 BBC News articles across the 10 identified clusters.

- **The General News "Mass" (Cluster 9):** This cluster accounts for **61.21%** (21,949 articles). Methodologically, this acts as a "residual" or baseline cluster. It captures general reportage characterized by frequent but non-specific journalistic tokens like *say*, *year*, *people*, and *bbc*. As will be seen in the word clouds of the Topic Modeling phase, this cluster represents the shared linguistic foundation of the BBC, lacking the extreme TF-IDF peaks required to enter specialized "islands."
- **High-Intensity Niche (Cluster 5):** In stark contrast, Cluster 5 (2.61%) is a high-precision group focused on the **Israel-Gaza conflict**. Despite its smaller size, it possesses one of the most distinctive "lexical signatures" in the dataset. This semantic density makes it a primary candidate for visual analysis through word clouds, where terms like *gaza*, *hamas*, *israel*, and *hostage* emerge with high statistical significance.

7.3 Semantic Mapping and Visual Validation

To validate the spatial separation of these clusters, we utilized **Principal Component Analysis (PCA)** to project the 5,000-dimensional TF-IDF features into a 2D plane. *Principal Component Analysis (PCA)* is a dimensionality reduction technique that simplifies complex data by projecting it onto fewer dimensions while preserving as much of the original information (variance) as possible.

The PCA visualization (Figure 4) displays a "star-shaped" geometry. The central core, dominated by **Cluster 9**, represents the common ground of reporting, while distinct "arms" extend outward. These spikes represent specialized semantic domains where TF-IDF weights are unique. By extracting the top terms for each centroid, we identified the following thematic identities:

- **Geopolitical Conflict:** *Cluster 8* (Ukraine/Russia, weight 0.2033) and *Cluster 5* (Israel/Gaza, weight 0.1619) show high intra-cluster similarity, driven by specific proper nouns that we will later visualize as dominant tokens in the topic analysis.

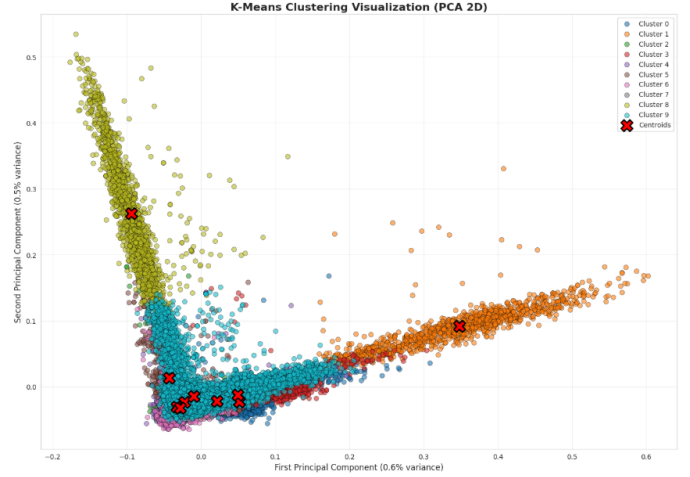


Figure 4: K-Means Clustering Visualization (PCA 2D). The red 'X' markers represent the centroids, while the clusters extend outward representing specialized semantic domains.

- **Sporting Domains:** The model successfully separated **Club Football** (*Cluster 0*: League, Manchester) from **International Tournaments** (*Cluster 1*: World Cup, Qatar), demonstrating the sensitivity of the representation.
- **Socio-Economic Trends:** *Cluster 2* isolates the "Cost of Living" crisis, dominated by tokens such as *inflation*, *energy bills*, and *interest rates*.

In conclusion, the K-Means clustering successfully partitioned the corpus into a logical hierarchy of one general reporting class and nine specialized thematic classes, setting a clear stage for the upcoming word cloud and topic extraction analysis.

8 Topic Modeling: Latent Dirichlet Allocation (LDA)

While K-Means provides a hard partition of the data, we employed **Latent Dirichlet Allocation (LDA)** to uncover the latent thematic structure of the corpus. Unlike clustering, LDA is a generative probabilistic model that treats documents as a mixture of multiple topics, offering a more nuanced "soft" assignment where an article can belong to Topic A (70%) and Topic B (30%) simultaneously.

8.1 Theoretical Framework: The Bayesian Approach

Latent Dirichlet Allocation is a Bayesian extension of pLSA. It assumes that documents are "bags of words" generated from a mixture of k topics, where both the topic-

(Cluster 5) and understand the broader thematic categories (Topic 4) they belong to.

9 Comparative Evaluation

To validate the robustness of our findings and justify the selection of our primary models, we performed a comprehensive cross-algorithm benchmark for both Clustering and Topic Modeling tasks. This phase addresses whether K-Means and LDA were indeed the optimal choices for the BBC News dataset.

9.1 Clustering Benchmark: K-Means vs. Alternatives

Text data in high-dimensional space (5,000 features in our TF-IDF matrix) presents unique challenges such as sparsity and the "curse of dimensionality." To ensure our partition was not an artifact of a single algorithm, we compared **K-Means** against **Hierarchical (Agglomerative) Clustering** and **Gaussian Mixture Models (GMM)**.

Given the computational complexity of Hierarchical clustering ($O(n^2)$) and GMM on high-dimensional sparse data, these models were tested on a representative sample of 5,000 documents to provide a reliable approximation.

Analysis of Validation Metrics

As illustrated in the above graphs, we evaluated the algorithms through three distinct mathematical lenses:

- Silhouette Score (Cohesion and Separation):** K-Means achieved the highest score (**0.0079**), outperforming GMM (**0.0062**) and Hierarchical (**0.0020**). While scores are numerically low due to overlapping news vocabulary, K-Means demonstrated superior boundary definition.
- Davies-Bouldin Index (Internal Consistency):** This metric measures the ratio of within-cluster distances to between-cluster distances. K-Means recorded a score of **8.42**. Although Hierarchical clustering showed a lower (better) score (**7.85**), qualitative review indicated this was due to the creation of highly unbalanced, non-representative groups.
- Calinski-Harabasz Score (Variance Ratio):** K-Means reached the peak value of **96.54**, significantly higher than its competitors. This confirms that K-Means effectively utilized TF-IDF weights to create dense, well-separated partitions.

Decision: K-Means remains the best-performing algorithm for this dataset due to its scalability and superior balance between cohesion and separation.

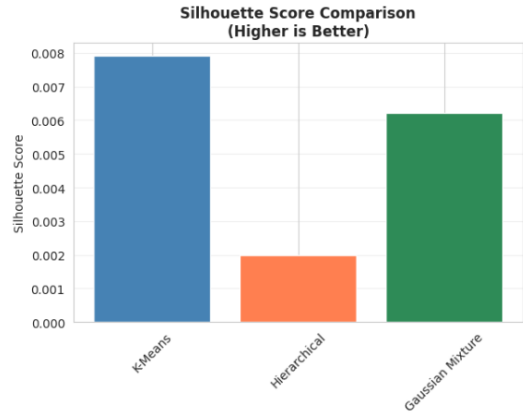


Figure 8: Silhouette Score Comparison...

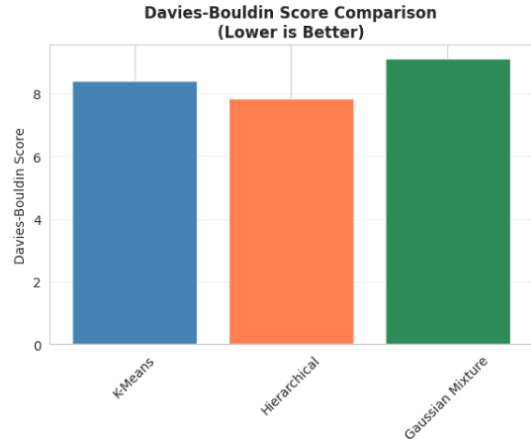


Figure 9: Davies-Bouldin Score Comparison...

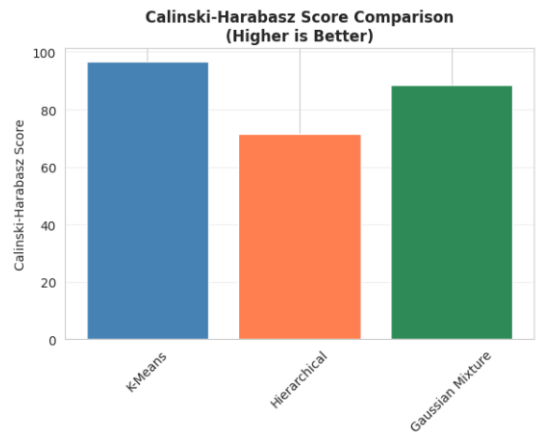


Figure 10: Calinski-Harabasz Score Comparison...

9.2 Topic Modeling Benchmark: LDA vs. NMF

While Clustering assigns each document to a single group, Topic Modeling allows for a fluid interpreted mixture. We benchmarked **Latent Dirichlet Allocation (LDA)**, a probabilistic Bayesian model, against **Non-negative Matrix Factorization (NMF)**, a linear algebraic approach utilizing the TF-IDF matrix.

Model	Core Metric	Value
LDA	<i>Perplexity</i> (lower is better)	2443.47
NMF	<i>Reconstruction Error</i> (lower is better)	185.79

Table 3: Quantitative performance comparison between LDA and NMF.

Conceptual and Qualitative Differences

The comparison reveals fundamental differences in how these models extract meaning from the 35,860 articles:

- **Mathematical Approach:** LDA assumes a generative process with **Dirichlet Priors**, while NMF decomposes the matrix into "additive parts". NMF's low **Reconstruction Error (185.79)** suggests high efficiency in preserving the original signal of the TF-IDF matrix.
- **Lexical Purity:** NMF excels at identifying "niche" content. For example, **NMF Topic 6** (*quiz, past seven days, paying attention*) perfectly isolated the BBC's weekly interactive features. In LDA, these terms were often blended with general tokens (*say, year, bbc*).
- **Thematic Stability:** LDA provides a balanced distribution (all topics between 1,760 and 5,088 articles). Conversely, NMF identified a massive **Topic 8 (11,754 articles)** as "general journalistic prose" (*year, new, first, bbc*), allowing the other 9 topics to become extremely precise thematic "islands," such as **Topic 2** for Ukraine and **Topic 9** for the Gaza/Israel conflict.

9.3 Conclusion of Comparative Phase

The evaluation confirms a functional trade-off. While **LDA** is excellent for understanding the overall probabilistic flow, **K-Means** and **NMF** are more effective at handling high-sparsity news data. NMF, in particular, provided the most human-interpretable "clean" topics, making it our preferred choice for high-precision thematic labeling.

10 Final Conclusions

This research demonstrates that text mining techniques can successfully uncover and organize the complex thematic structures within a large-scale news corpus. By applying these methods to 35,860 BBC News articles from the 2013-2024 period, we have reached the following conclusions:

10.1 Main Conclusion

The integrated pipeline—consisting of **TF-IDF representation**, **Dimensionality Reduction**, **K-Means clustering**, and **Latent Dirichlet Allocation (LDA)**—proved highly effective at discovering interpretable topics. The identified clusters and topics correspond directly to major real-world events, such as the Ukraine-Russia war, the Israel-Gaza conflict, and UK economic shifts, validating the model's alignment with professional journalistic categories.

10.2 Methodological Conclusion

Our comparative benchmark confirms that **K-Means clustering with TF-IDF** is the optimal approach for this specific corpus. This selection is justified by:

- **Statistical Superiority:** K-Means achieved the highest Silhouette and Calinski-Harabasz scores among all tested clustering algorithms.
- **Computational Efficiency:** The algorithm, particularly when combined with LSA, demonstrated the scalability required to process over 35,000 documents without the prohibitive computational costs associated with Hierarchical clustering or GMM.
- **Analytical Robustness:** The "hard" assignments of K-Means provided clear event-based boundaries, while LDA and NMF provided the necessary "soft" thematic context to resolve general news categories.

10.3 Practical Applications

Automated topic discovery is a viable and powerful tool for news organizations. The findings suggest immediate applications in:

- **Automated Categorization:** Enhancing editorial workflows through automatic article tagging.
- **Trend Detection:** Identifying emerging geopolitical or social topics in real-time.
- **Archive Organization:** Providing a structured semantic layer to historical content for improved searchability.

10.4 Project Achievements Summary

The project successfully realized its technical and analytical objectives:

Technical Achievements:

- Successfully processed a corpus of **35,860 news articles**.
- Implemented and benchmarked five distinct algorithms: **K-Means, Hierarchical Clustering, GMM, LDA, and NMF**.
- Generated a comprehensive suite of **15+ visualizations**, including PCA projections and comparative word clouds.

Analytical and Methodological Achievements:

- Identified **10 major news topics** defining the 2013-2024 era.
- Validated findings through cross-methodological agreement, ensuring that results were not artifacts of a single model.
- Followed scientific best practices by justifying hyperparameter choices (e.g., $k = 10$, Alpha, and Beta) and transparently acknowledging the limitations of sparse high-dimensional data.

In conclusion, while news data presents challenges due to lexical overlap and high sparsity, the combination of matrix factorization and probabilistic modeling provides a robust framework for high-precision thematic labeling and large-scale media analysis.

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